Microsoft Elevate

CAPSTONE PROJECT

E-Commerce Sales Analysis Using Power BI

PRESENTED BY

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OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

E-commerce platforms generate large volumes of transactional and customer data every day. However, organizations often face challenges in analyzing this data effectively to understand sales performance, customer purchasing behavior, and product demand patterns.

Without proper data analysis tools, it becomes difficult for businesses to identify profitable products, monitor regional sales performance, and make data-driven decisions to improve business strategies.

PROPOSED SOLUTION:

The proposed system uses **Microsoft Power BI** to analyze and visualize e-commerce sales data. The system will create interactive dashboards that help businesses understand sales trends and customer behavior.

Key objectives include:

- Analyze total revenue and sales trends
- Identify top-selling products
- Understand customer purchasing behavior
- Analyze region-wise sales performance
- Provide interactive dashboards for business insights

SYSTEM APPROACH:

Tools and Technologies

- Microsoft Power BI Desktop
- Excel / CSV Dataset
- Power Query for Data Cleaning
- Data Modeling in Power BI

System Requirements

- Windows OS
- Power BI Desktop
- E-commerce dataset

Libraries / Features Used

- Power Query Editor
- DAX (Data Analysis Expressions)
- Visualization tools (charts, maps, tables)

ALGORITHM & DEPLOYMENT:

Data Input

- The dataset includes:
- Order ID
- Product name
- Category
- Sales amount
- Quantity
- Customer location
- Order date

Data Processing Steps

- **Data Collection**
  E-commerce sales dataset collected from open data sources.
- **Data Cleaning**
  Handling missing values and removing duplicate records.
- **Data Transformation**
  Converting data types and creating calculated fields.
- **Data Visualization**
  Creating dashboards using charts and graphs.

Deployment

- The Power BI dashboard is deployed locally using **Power BI Desktop**, enabling interactive data exploration and reporting.

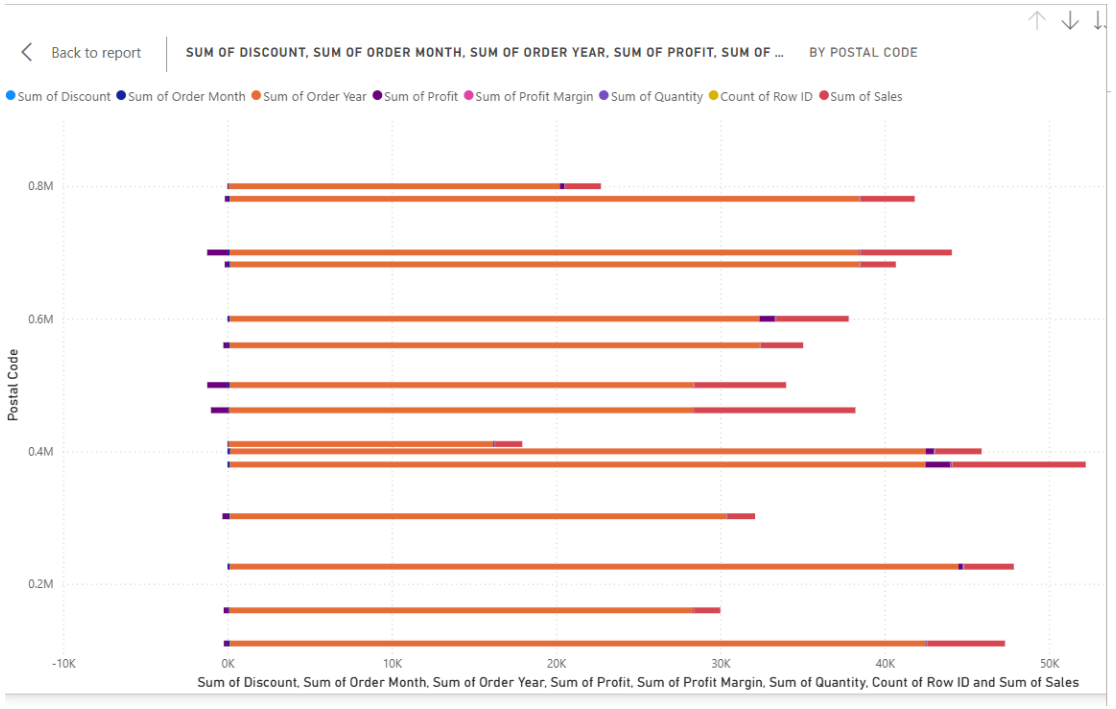
RESULT:

The Power BI dashboard provides insights such as:

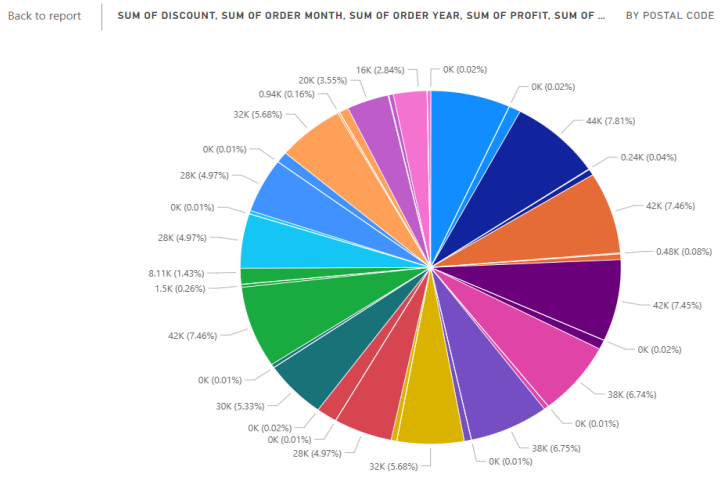
- Total Sales Revenue
- Top 10 Best-Selling Products
- Monthly Sales Trend
- Region-Wise Sales Distribution
- Category-Wise Performance

Visualizations used:

- Bar charts
- Pie charts
- Line charts
- Map visualization
- KPI cards



RESULT: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BY POSTAL CODE
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0K (0.02%)
20K (3.55%)
0K (0.02%)
0.94K (0.16%)
32K (5.68%)
44K (7.81%)
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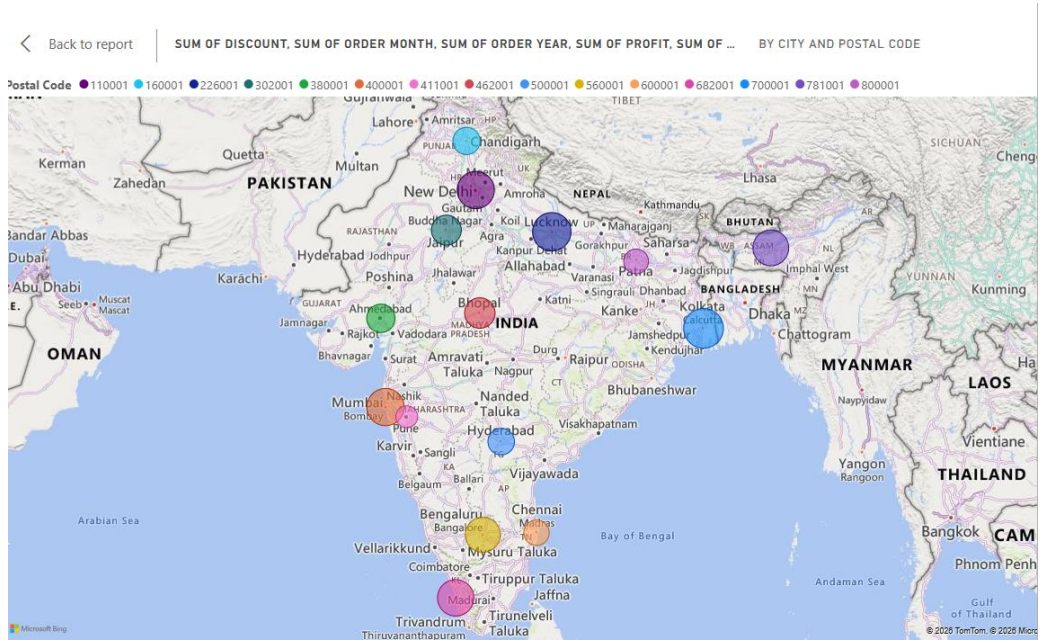
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BY CITY AND POSTAL CODE
Postal Code 110001 160001 226001 302001 380001 400001 411001 462001 500001 560001 600001 682001 700001 781001 800001
PAKISTAN
INDIA
NEPAL
BHUTAN
BANGLADESH
MYANMAR
LAOS
THAILAND
OMAN
Arabian Sea
Bay of Bengal
Andaman Sea
Gulf of Thailand
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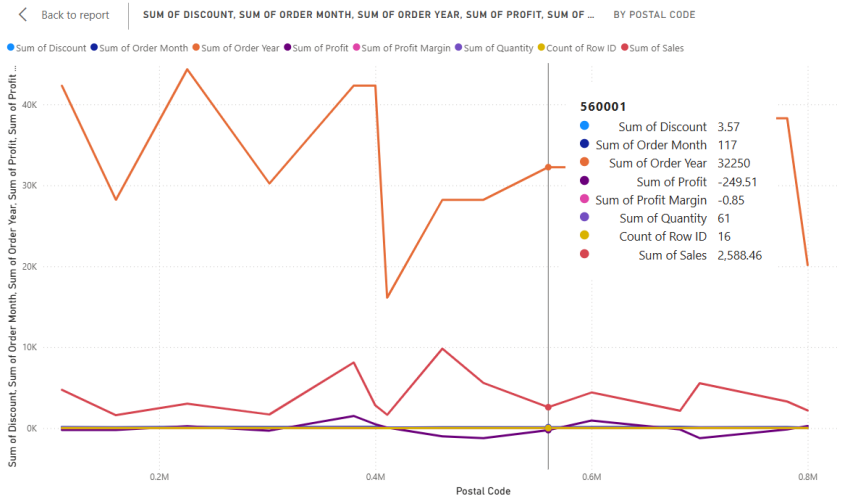
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SUM OF DISCOUNT, SUM OF ORDER MONTH, SUM OF ORDER YEAR, SUM OF PROFIT, SUM OF ...
BY POSTAL CODE
Sum of Discount
Sum of Order Month
Sum of Order Year
Sum of Profit
Sum of Profit Margin
Sum of Quantity
Count of Row ID
Sum of Sales
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Sum of Discount 3.57
Sum of Order Month 117
Sum of Order Year 32250
Sum of Profit -249.51
Sum of Profit Margin -0.85
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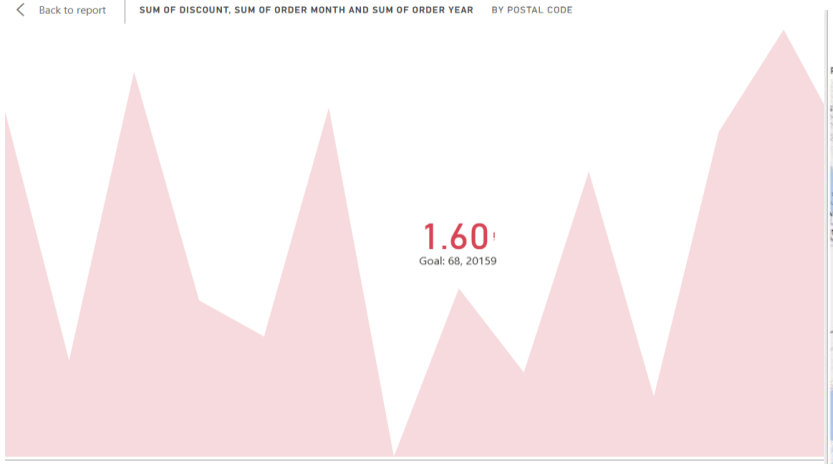
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BY POSTAL CODE
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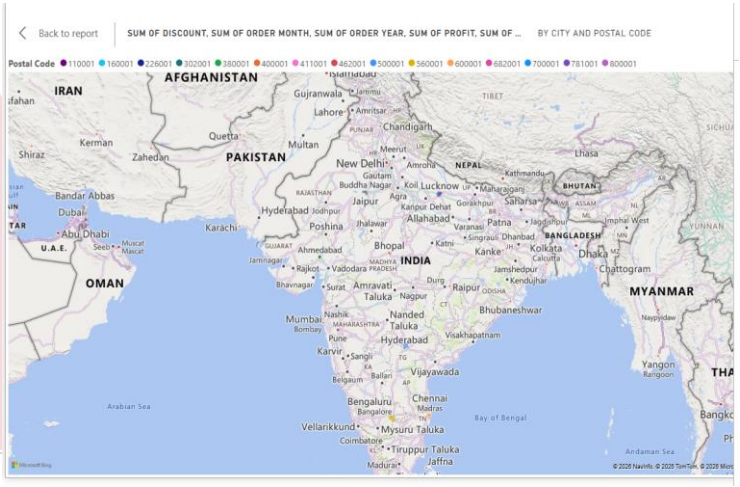
Goal: 68, 20159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SUM OF DISCOUNT, SUM OF ORDER MONTH, SUM OF ORDER YEAR, SUM OF PROFIT, SUM OF ...
BY CITY AND POSTAL CODE
Postal Code 110001 160001 226001 302001 380001 400001 411001 462001 500001 560001 600001 682001 700001 781001 800001
IRAN
AFGHANISTAN
PAKISTAN
INDIA
NEPAL
BHUTAN
BANGLADESH
MYANMAR
OMAN
U.A.E.
Arabian Sea
Bay of Bengal
Andaman Sea

CONCLUSION:

The E-Commerce Sales Analysis system successfully demonstrates how Power BI can transform raw sales data into meaningful business insights. The interactive dashboards help businesses understand customer behavior, monitor sales trends, and identify high-performing products.

This project highlights the importance of data visualization in improving decision-making and business strategy.

FUTURE SCOPE:

Future enhancements for this system include:

- Real-time sales data integration
- AI-based sales prediction
- Customer recommendation systems
- Integration with cloud platforms such as Azure

REFERENCES:

- Microsoft Power BI Documentation
- Kaggle E-commerce Datasets
- Data Visualization Best Practices
- Online research articles related to sales analytics

GitHub Link: Link

Thank You